Atemporality, Emergent Time, and the Born Rule from Partial Descriptions in Information Geometry

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Abstract. We develop an information-geometric framework in which the absence and emergence of temporal structure can be analyzed within a unified formal setting. We distinguish between complete descriptions of physical systems, modeled as maximal statistical manifolds satisfying conditions of statistical sufficiency, reparametrization invariance, and statistical locality, and partial descriptions associated with internal observers.

Under an explicit symmetry assumption (Assumption A1) on the space of partial descriptions, we show that any complete description is atemporal: no preferred ordering parameter survives averaging over all observers. Conversely, for partial descriptions updated through successive interactions, we introduce an accumulated parameter τ arising from information-theoretic asymmetries, and show that it converges under suitable mixing conditions to a continuous stochastic process with the formal properties expected of physical time.

In the quantum setting, we analyze functionals compatible with the Bures metric under conditions of positivity, normalization, linearity, and monotonicity under completely positive trace-preserving maps. We prove that such functionals reduce to Gleason-type measures on the lattice of projectors, thereby recovering the Born rule within this framework.

We emphasize that the connection to canonical quantum gravity is structural rather than constructive: both the present framework and the Wheeler–DeWitt equation exhibit the absence of intrinsic time at the level of complete descriptions, while temporal ordering emerges only for subsystems. The results suggest that aspects of temporal structure and quantum probability may be understood as consequences of information-geometric consistency, although several foundational and dynamical questions remain open.

1. Introduction

The problem of time in quantum gravity and the foundational status of the Born rule are two of the most persistent open questions in theoretical physics. The Wheeler–DeWitt equation [1, 2], which governs the quantum state of the universe in canonical quantum gravity, contains no explicit time parameter. This absence is not a technical artifact but a structural feature: the Hamiltonian constraint annihilates all physical states, leaving no room for a preferred temporal direction. Simultaneously, the Born rule — which assigns probabilities to measurement outcomes — has resisted derivation from more fundamental principles despite numerous attempts [3, 4, 5, 6].

In this paper we propose that both features are consequences of a single structural property of information-geometric descriptions of physical systems. We work within the framework of statistical manifolds equipped with the Fisher information metric, and we build on three classical results: Čencov's theorem [7], which establishes the uniqueness of the Fisher metric under natural invariance conditions; the quantum extension by Petz [8], which characterizes the family of monotone metrics on the space of density matrices; and the Löwner–Heinz theorem [9], which ensures operator monotonicity of the relevant functional classes.

Our main contributions are three. First, we show that, *under an explicit reflection symmetry assumption (A1)* on the space of partial descriptions, complete descriptions are atemporal in the sense that the expected temporal parameter over all observers vanishes exactly (Theorem 1). Second, we identify a parameter τ that emerges from partial descriptions and satisfies the formal properties of physical time under coarse-graining (Theorem 2). Third, we demonstrate that functionals compatible with the Bures metric under conditions X1–X4 reduce to Gleason-type measures, recovering the Born rule within this framework (Theorem 3).

We emphasize what we do not claim. Assumption A1 is an explicit symmetry condition, not derived from C1–C3; the scope of Theorem 1 is conditional on it. The connection to Wheeler–DeWitt is a structural parallel, not a constructive embedding. In the derivation of Born for degenerate observables, Gleason's theorem is a central ingredient. These limitations define open problems for future work.

The paper is organized as follows. Section 2 introduces the mathematical framework. Section 3 establishes the atemporality theorem. Section 4 derives the emergence of time for partial observers. Section 5 derives the Born rule via the uniqueness theorem. Section 6 discusses the structural parallel with Wheeler–DeWitt. Section 7 presents experimental predictions. Section 8 discusses open problems.

2. Mathematical Framework

2.0. Notation and Conventions

We use the following conventions consistently throughout. The classical and quantum settings are treated in parallel; when a statement applies to both, we write it once and indicate (classical / quantum) where the objects differ.

Classical setting: the state space is a smooth family of probability distributions $S = \{p(x|\theta) : \theta \in \Theta \subset \mathbb{R}^n\}$. The parameter space Θ carries the Fisher information metric $g_{ij}(\theta)$. The complete description G is a terminal object in the category of statistical models satisfying conditions C1–C3 (see Section 2.2). A partial description P(o) is a k-dimensional submanifold of G with $k < N$. All divergences are classical KL divergences: $D_{kl}(p \parallel q) = \int p \log(p/q)\, dx$.

Quantum setting: the state space is the manifold of density matrices $D(H) = \{\rho \in B(H) : \rho \geq 0, \mathrm{Tr}(\rho) = 1\}$ on a Hilbert space H of dimension n. The metric is one of the Petz family of monotone

metrics, with the Bures metric as the canonical choice. The complete description G corresponds to the full D(H); a partial description P(o) corresponds to a submanifold of D(H) accessible to observer o, identified with the reduced density matrix $\rho_o = \mathrm{Tr}_{env}(\rho_{total})$. Divergences in the quantum setting are quantum relative entropies $S(\rho \square \sigma) = \mathrm{Tr}(\rho(\log \rho - \log \sigma))$, or Bures distances $D^{Bures}(\rho, \sigma) = 2(1 - F(\rho,\sigma))$ where $F(\rho,\sigma) = \mathrm{Tr}(\sqrt{(\sqrt{\rho}\, \sigma \sqrt{\rho})})$.

Shared notation: G denotes the complete description in both settings. P(o) denotes the partial description of observer o. $\sigma(o) \in \{+1, -1\}$ is the binary anomaly variable defined in Section 4. τ is the emergent temporal parameter. Conditions C1–C3 apply to both settings. Conditions X1–X4 in Section 5 are stated exclusively in the quantum setting. When we write $\mathrm{Tr}(\cdot)$ we mean the operator trace in the quantum setting and the integral $\int \cdot\, dx$ in the classical setting.

2.1. Statistical Manifolds and the Fisher Metric

Let X be a measurable space and let $S = \{p(x|\theta) : \theta \in \Theta\}$ be a smooth family of probability distributions parametrized by $\theta \in \Theta \subseteq \mathbb{R}^n$. The Fisher information metric is the Riemannian metric on Θ defined by

$$g_{ij}(\theta) = E_\theta[\partial_i \log p(x|\theta) \cdot \partial_j \log p(x|\theta)]$$

where $\partial_i = \partial/\partial\theta^i$ and the expectation is taken with respect to $p(x|\theta)$. The pair (Θ, g) constitutes a statistical manifold.

2.2. The Complete Description G and Conditions C1–C3

We define the complete description **G** as the maximal statistical model associated with a physical system, understood as a statistical manifold that is closed under sufficient statistics and admits no non-trivial coarse-graining that preserves the Fisher metric. Formally, G is characterized as a *terminal object* in the category of statistical models satisfying conditions C1–C3 below. A partial description P(o) is a k-dimensional submanifold of G with $k < N = \dim(G)$.

Condition C1 (Statistical Sufficiency). Two states of the system are identified in D if and only if they are statistically indistinguishable. In the quantum case, this corresponds to monotonicity under completely positive trace-preserving (CPTP) maps.

Condition C2 (Reparametrization Invariance). The description is invariant under smooth reparametrizations $\theta \to \theta'(\theta)$. In the quantum case, this is unitary invariance: $g^U\rho^U\dagger(UAU\dagger, UBU\dagger) = g\rho(A, B)$ for all unitary U.

Condition C3 (Statistical Locality). For independent subsystems with states $\rho^A \square \rho^B$, the metric of the composite system is the direct sum of the individual metrics. In the quantum case, this corresponds to additivity on product states.

The verification that these three conditions have exact quantum analogues—not merely analogies—was established in [8, 9] and is recalled in the Appendix.

2.3. The Information Distance Δ(o)

We define the information distance between the complete description and a partial description as the projection divergence:

$$\Delta(o) := \inf_{q \in P(o)} D_{KL}(p_G \,/\!/\, q)$$

where p^G denotes the reference distribution induced by the complete model G. When P(o) is a closed submanifold of G, this infimum is attained at the unique closest point under the Fisher metric, and $\Delta(o) \geq 0$ by non-negativity of KL divergence. This definition replaces the informal $\Delta(o) = D_{KL}(G \,\|\, P(o))$ of earlier drafts, which was not well-posed as G and P(o) are manifolds rather than individual distributions.

2.4. The Bures Metric and Quantum Extension

In the quantum setting, the role of the Fisher metric is played by the family of monotone metrics characterized by Petz [8]. Among these, the Bures metric occupies a distinguished position as the minimal monotone metric and extends continuously to pure states. For a density matrix ρ and perturbations A, B, the Bures metric is

$$g^{Bures}_{\rho}(A, B) = Tr(A \cdot J_{\rho}^{-1}(B))$$

where $J_\rho(X) = (\rho X + X\rho)/2$ is the symmetric Jordan product. The Bures distance between two states ρ and σ is $D^{Bures}(\rho, \sigma) = 2(1 - F(\rho, \sigma))$ where $F(\rho, \sigma) = Tr(\sqrt{(\sqrt{\rho}\, \sigma \sqrt{\rho})})$ is the fidelity.

3. The Atemporality Theorem

3.1. Atemporality under Symmetry of Partial Descriptions

We formalize the notion of atemporality as the absence of a preferred ordering parameter compatible with the information geometry of the complete description.

Let G denote the complete statistical manifold of a physical system as defined in Section 2.2. Let Gr(k,N) denote the space of admissible k-dimensional partial descriptions $P(o) \subset G$, equipped with a measure dμ(o) induced by the restriction of the information metric to each P(o).

> **Principle of Global Informational Neutrality.** A complete description of a closed physical system should not privilege any observer-dependent direction of information accumulation. Any temporal orientation appearing in a partial description must arise from conditioning, coarse-graining, or boundary data, not from the complete description itself.

Assumption A1 below is the mathematical implementation of this principle.

Before stating A1, we provide its physical motivation. Since G represents the maximal description of the system, it does not correspond to the perspective of any particular observer. Consequently, any direction of information gain defined by one partial description must be paired with an opposite direction associated with another admissible partial description, unless a global boundary condition explicitly breaks this symmetry. This is analogous to equilibrium ensembles in statistical mechanics: at the microscopic level, individual trajectories may exhibit directed behavior, but the full equilibrium ensemble does not select a preferred direction in phase space. Directional bias

appears only after conditioning on a macrostate, a boundary condition, or a restricted subsystem. In the present framework, partial descriptions P(o) play the role of such conditioned subsystems.

Assumption A1 is therefore not a dynamical law but a global neutrality condition. It expresses the claim that, before conditioning on a particular observer, the complete description G contains no intrinsic distinction between past-directed and future-directed informational orientations. Time arises only after this symmetry is broken by restriction to a partial description. This interpretation also clarifies when A1 may fail: if the universe possesses global boundary conditions that select a preferred informational orientation — such as a special low-entropy initial condition, asymmetric cosmological boundary data, or a global constraint accessible to all observers — then the involution R may fail to be measure-preserving. In that case Theorem 1 would no longer apply in its symmetric form.

Formally, A1 is an additional structural condition on the space of observers, *not derived from C1–C3*, and should be understood as the information-geometric analogue of a no-preferred-frame principle: the complete description is temporally neutral, while temporal direction emerges from conditioning on partial descriptions.

Assumption A1 (Reflection Symmetry). There exists a measurable involution $R : Gr(k,N) \to Gr(k,N)$ such that:

(i) $R(R(o)) = o$ for all o,

(ii) $d\mu(R(o)) = d\mu(o)$ (measure-preserving),

(iii) $\sigma(R(o)) = -\sigma(o)$ (sign reversal of the anomaly variable),

(iv) $\Delta(R(o)) = \Delta(o)$ (invariance of the information distance).

Theorem 1 (Atemporal Expectation under Symmetry). Under Assumption A1, the expectation value of the emergent temporal parameter

$$\tau(o) := \sigma(o) \cdot \Delta(o) / \sqrt{k}$$

over the space of observers vanishes:

$$\mathbb{E}_o[\tau] = 0.$$

Proof.

By definition:

$$\mathbb{E}_o[\tau] = [\int_{Gr(k,N)} \sigma(o)\Delta(o) \, d\mu(o)] / [\int_{Gr(k,N)} d\mu(o)].$$

Applying the change of variables $o \mapsto R(o)$ in the numerator, and using properties (ii), (iii), (iv) of A1:

$$\int \sigma(o)\Delta(o) \, d\mu(o) = \int \sigma(R(o))\Delta(R(o)) \, d\mu(R(o)) = \int [-\sigma(o)]\Delta(o) \, d\mu(o).$$

Therefore the integral equals its own negative, so it vanishes. Hence $\mathbb{E}_o[\tau] = 0$. □

Interpretation and Status of Assumption A1. Assumption A1 should be understood as an idealized symmetry condition at the level of the complete description. It expresses the absence of a preferred global direction of information flow when all admissible partial descriptions are treated on equal footing.

More precisely, A1 encodes the requirement that the space of observers does not privilege any orientation of informational deviation relative to the complete model G. In this sense, it is analogous to equilibrium symmetry conditions in statistical mechanics, where macroscopic directional biases vanish under complete averaging.

We emphasize that A1 is not derived from conditions C1–C3. Rather, it is an additional structural assumption reflecting the absence of a global informational arrow in the complete description. Determining whether A1 can be derived from dynamical laws, cosmological boundary conditions, or typicality arguments remains an open problem.

Remarks on Physical Realizability. Assumption A1 is expected to hold in idealized situations where the space of admissible observers is sufficiently rich and symmetric, for instance in ergodic or equilibrium-like regimes. It may fail in the presence of global constraints, such as special initial conditions or boundary-induced asymmetries. In such cases, the conclusion of Theorem 1 need not hold, and residual temporal bias at the level of the complete description may persist.

3.2. Typicality Argument for Assumption A1

We give a conditional typicality argument motivating Assumption A1. The goal is not to prove A1 from first principles, but to show that A1 is the generic expectation for complete descriptions whose admissible observer subspaces are sampled without a globally preferred informational orientation.

Proposition (Typicality Implies Reflection Symmetry). Let G be a complete statistical manifold equipped with an information-geometric measure $d\mu$. Suppose that:

(i) admissible observers are typical k-dimensional coarse-grainings of G,

(ii) the induced measure on Gr(k,N) is invariant under reversal of informational orientation ($\sigma \to -\sigma$),

(iii) the information distance $\Delta(o)$ depends only on the magnitude of deviation from G, not on its orientation.

Then Assumption A1 holds up to a set of measure zero.

Proof. By (ii), for almost every P(o) with orientation $\sigma(o)$, there exists P(o') with orientation $-\sigma(o)$ and equal measure $d\mu(o') = d\mu(o)$. By (iii), $\Delta(o') = \Delta(o)$. Defining $R(o) = o'$ gives a measurable involution satisfying $R(R(o)) = o$, $d\mu(R(o)) = d\mu(o)$, $\sigma(R(o)) = -\sigma(o)$, and $\Delta(R(o)) = \Delta(o)$. These are exactly the conditions of A1. □

Remark. We note that hypothesis (ii) is itself a symmetry condition; the Proposition reduces A1 to the assumption that the sampling of observers is informationally unbiased.

This invariance should be understood as a typicality assumption rather than a symmetry imposed at the level of individual observers: it asserts that no observer orientation is preferred in the sampling measure as a generic property of the ensemble, not as a constraint on each element. This may be more physically transparent even if not strictly weaker than A1. Hypothesis (ii) would fail if the space of observers were generated by a process with a built-in directional bias, such as a universe with a special low-entropy initial condition selecting a preferred informational orientation for a large class of coarse-grainings.

Observation 1 (Structural Parallel with Wheeler–DeWitt). The Wheeler–DeWitt equation $\hat{H}|\Psi\rangle = 0$ admits no intrinsic temporal parameter. This is a structural parallel with Theorem 1, not a constructive embedding. The superspace metric G_{ijkl} of Wheeler–DeWitt has indefinite signature, while the Fisher metric is positive semi-definite; these are mathematically distinct objects. Both exhibit absence of preferred time under complete descriptions, and in the Gaussian approximation for the space of 3-metrics the two coincide locally. We use the phrase 'structural parallel' deliberately: we do not construct a functor between the respective categories. No results of this paper depend on this structural parallel.

4. Emergent Time from Partial Descriptions

4.1. The Binary Anomaly Variable σ

Let G be the complete statistical manifold and let P(o) be the k-dimensional submanifold accessible to observer o, with k < N. We define the anomaly of P(o) with respect to G as

$$\Delta(o) = \inf_{q \in P(o)} D_{KL}(p_G \,/\!/\, q) \geq 0$$

We define the binary anomaly variable dynamically by comparing successive information distances. We denote by $\sigma(o_n)$ the anomaly variable associated with the n-th update, and write $\sigma_n := \sigma(o_n)$ for notational convenience. Given two consecutive partial descriptions $P(o_n)$ and $P(o_{n+1})$, set

$$\sigma_n := \mathrm{sgn}(\Delta(o_n) - \Delta(o_{n+1})).$$

Thus $\sigma_n = +1$ when the observer's partial description moves closer to the complete description G (information gain), and $\sigma_n = -1$ when it moves farther away (information loss or fluctuation). This definition replaces the informal description of σ as the sign of a deviation along a maximal information gradient, and makes the anomaly variable explicitly dynamical. The variable σ_n is a geometric quantity derived from the structure of the statistical manifold, not a probability.

4.2. Emergence of Continuous Time as a Consistency Result

The emergence of a temporal parameter in this framework should be understood as a consistency result rather than a derivation from first-principles dynamics. Specifically, we show that if the sequence of informational updates associated with a partial observer satisfies standard stochastic conditions, then a parameter with the formal properties of physical time necessarily arises.

We do not derive the stochastic properties of the sequence $\{\sigma_n, \Delta_n\}$ from an underlying dynamical theory. Instead, we identify sufficient conditions under which temporal behavior emerges, leaving their dynamical origin as an open problem.

Theorem 2 (Emergent Time under Stochastic Consistency Conditions). Let $\{\sigma_n\}_{n=1}^{k}$ be a sequence of binary anomaly variables associated with successive updates of a partial description P(o), and let $\Delta_n = \Delta(o_n)$ denote the corresponding information distances. Assume:

(i) $^{O}[\sigma_n] = \mu$, $\mathrm{Var}(\sigma_n) = 1$,

(ii) the sequence satisfies a mixing condition: $\Sigma_j |\mathrm{Cov}(\sigma_n, \sigma_{n+j})| < \infty$,

(iii) the information distances satisfy $^{O}[\Delta_n] < \infty$ and $^{O}[\Delta_n^2] < \infty$.

Define the accumulated parameter:

$$\tau(k) = (1/\sqrt{k}) \cdot \Sigma^{n} \sigma\square \cdot \Delta\square.$$

Then, as $k \to \infty$:

(a) $\tau(k)$ converges in distribution to a Gaussian random variable,

(b) $\tau(k)$ defines a continuous stochastic process in the scaling limit,

(c) when $\mu \neq 0$, τ exhibits a monotonic drift, defining a preferred direction.

The mixing condition (ii) can be replaced by any standard sufficient condition ensuring convergence to the central limit theorem, such as strong mixing (α-mixing) or martingale difference conditions. The specific growth condition $D_KL(G \parallel P(o_{n+j})) - D_KL(G \parallel P(o_n)) \geq c \cdot j\alpha$ for $c > 0$, $\alpha > 1/2$ is one explicit sufficient example rather than a necessary requirement. Note that (iii) is non-trivial only for Δ_n; the condition $\sigma_n \in \{+1,-1\}$ ensures bounded moments for the binary component automatically.

Interpretation. The theorem shows that temporal structure emerges whenever the sequence of informational updates exhibits a persistent statistical bias and satisfies standard regularity conditions. The parameter τ is therefore not postulated but arises as the natural scaling limit of accumulated informational asymmetries. Importantly, the result is conditional: it does not explain why the underlying sequence satisfies these properties. Instead, it establishes that, given such conditions, the emergence of a time parameter is unavoidable.

4.3. Physical Interpretation

The information-geometric objects introduced above have the following physical interpretations:

$\sigma(o)$: the direction of information update. A positive value indicates the observer's description is gaining information relative to the complete model; negative indicates the reverse. It is the sign of the asymmetry in the current observation.

τ: the accumulated statistical bias of information exchanges. Formally, $\tau(k) = (1/\sqrt{k})\Sigma_n \sigma_n\Delta_n$; this is the signed sum of information updates, rescaled. When $\mu > 0$, the observer

consistently gains information and τ increases monotonically, defining a preferred direction. This is the arrow of time within the framework.

Time in this framework is the statistical bias in the observer's learning process relative to the complete description G. It is not a property of G itself — which is symmetric under $\sigma \to -\sigma$ by Theorem 1 — but of the partial description P(o) under successive interactions. The transition from the atemporal complete description to experienced time corresponds to the symmetry breaking from G to P(o).

4.4. The Direction of Time

The bias $\mu = {}^{O}[\sigma_n]$ represents the systematic asymmetry of the observer's anomalies. When $\mu > 0$, the observer systematically acquires information — τ increases monotonically and defines a preferred direction. This asymmetry follows from the irreversibility of information accumulation: once P(o) has updated to incorporate new information, the prior state of P(o) is not recoverable without additional information. The arrow of time is therefore not postulated but is the statistical consequence of the entanglement between the observer and the system under coarse-graining.

4.5. Scope and Limitations

A full derivation of the stochastic properties assumed in Theorem 2 from an underlying physical dynamics — such as unitary evolution combined with coarse-graining — remains an open problem. In this sense, the present result should be viewed as identifying a universality class of informational processes that give rise to temporal behavior, rather than as a complete dynamical theory of time. The value of the result lies in showing that temporal emergence is not sensitive to the specific microscopic dynamics: any process satisfying the regularity conditions (i)–(iii) will produce a time parameter with the same macroscopic properties.

4.6. A Toy Dynamical Model for the Anomaly Variable

We give a simple dynamical model showing how the binary anomaly variable σ_n may arise from repeated observer-system interactions, thereby motivating the stochastic assumptions of Theorem 2. This toy model is not intended as a fundamental dynamical theory, but as an existence proof that the anomaly variable σ_n can arise from simple update dynamics.

Consider an observer updating a partial description $P(o_n)$ at each interaction step n. Model the update of the information distance by

$$\Delta_{n+1} = \Delta_n - \eta I_n + \xi_n$$

where $\eta > 0$ is a learning rate, $I_n \geq 0$ is the information gained at step n, and ξ_n is a bounded fluctuation term representing noise, backreaction, or imperfect coarse-graining. Then

$$\sigma_n = \mathrm{sgn}(\Delta_n - \Delta_{n+1}) = \mathrm{sgn}(\eta I_n - \xi_n).$$

Thus σ_n is not imposed externally but is generated by the competition between systematic information gain and stochastic fluctuations. When the observer is learning on average (${}^{O}[\eta I_n - \xi_n] > 0$), we get $\mu = {}^{O}[\sigma_n] > 0$ and a positive temporal drift emerges. When the observer is in

informational equilibrium ($^{O}[\eta\ I_n - \xi_n] = 0$), positive and negative updates occur with equal probability and no preferred temporal direction emerges. If the noise ξ_n has finite correlation length, or if the information gains I_n are generated by weakly correlated interactions, the mixing condition (ii) of Theorem 2 is satisfied automatically.

Quantum Interpretation. In the quantum setting, the update rule can be motivated by repeated weak interactions between a system S, an observer O, and an environment E. Let the joint state evolve as $\rho_{SOE}^{(n+1)} = U_n\ \rho_{SOE}^{(n)}\ U_n^\dagger$, followed by coarse-graining over environmental degrees of freedom: $\rho_O^{(n+1)} = Tr_{SE}(U_n\ \rho_{SOE}^{(n)}\ U_n^\dagger)$. The information distance is then $\Delta_n = D_{Bures}(\rho_G, \rho_O^{(n)})$, and $\sigma_n = sgn(\Delta_n - \Delta_{n+1})$. This construction shows that σ_n can be generated by ordinary unitary dynamics at the global level combined with partial tracing at the observer level. Irreversibility is not introduced into the global dynamics; it appears only after restriction to a partial description.

5. The Born Rule from Gleason-Type Measures

5.1. Setup: Measurement as Entanglement

Let S be a quantum system in state $|\psi\rangle = \Sigma_i\ \alpha_i\ |i\rangle$ and let A be a measurement apparatus initially in state $|A_0\rangle$. A measurement interaction produces the entangled state

$$|\Psi_{SA}\rangle = \Sigma_i\ \alpha_i\ |i\rangle \otimes |A_i\rangle$$

where $\{|A_i\rangle\}$ are orthonormal pointer states ($\langle A_i|A_j\rangle = \delta_{ij}$). Tracing over S yields the reduced density matrix of the apparatus

$$\rho_A = Tr_S(|\Psi_{SA}\rangle\langle\Psi_{SA}|) = \Sigma_i\ |\alpha_i|^2\ |A_i\rangle\langle A_i|$$

5.2. The Uniqueness Theorem for the Temporal Bias Functional

Theorem 3 (Born Rule via Gleason-Type Measures). Let $\mu_i : (\rho_A, \Pi_i) \to \mathbb{R}$ be a functional assigning to each outcome i a temporal bias, where $\{\Pi_i\}$ is a complete set of orthogonal projectors. Suppose μ_i satisfies:

X1 (Positivity). $\mu_i(\rho, \Pi) \geq 0$ for all ρ, Π.

X2 (Normalization). $\Sigma_i\ \mu_i(\rho, \Pi_i) = D_{Bures}(\rho_A, \rho_{ref})$ for all ρ, $\{\Pi_i\}$.

X3 (Linearity in ρ). μ_i is linear in ρ_A.

X4 (Monotonicity under CPTP maps). For any CPTP map Λ: $\mu_i(\Lambda(\rho_A), \Lambda(\Pi_i)) \leq \mu_i(\rho_A, \Pi_i)$.

Then the assignment of measurement weights reduces to the class of Gleason-type measures on the lattice of projectors, and the unique functional satisfying X1–X4 is

$$\mu_i(\rho_A, \Pi_i) = D \cdot Tr(\rho_A \cdot \Pi_i)$$

where $D = D_{Bures}(\rho_A, \rho_{ref})$ is the total anomaly.

Proof. By X3 and the Riesz–Fréchet representation theorem, X3 implies $\mu_i(\rho_A, \Pi_i) = Tr(\rho_A \cdot M_\{\Pi_i\})$ for some positive Hermitian operator $M_\{\Pi_i\}$. By X2, $\Sigma_i M_\{\Pi_i\} = D \cdot I$. For non-degenerate observables (all Π_i rank-1), orthogonality $\Pi_i \Pi_j = 0$ implies $M_\{\Pi_i\} = D \cdot \Pi_i$ by algebraic structure of positive operators summing to a multiple of the identity with mutually orthogonal supports. For degenerate observables, condition X4 ensures σ-additivity of the functional on orthogonal decompositions, activating Gleason's theorem [10] as a central ingredient: any σ-additive measure on the lattice of closed subspaces of a Hilbert space of dimension ≥ 3 is of the form $Tr(\rho \cdot \Pi)$. The result $M_\{\Pi_i\} = D \cdot \Pi_i$ follows in full generality. □

5.3. Recovery of the Born Rule

Given $\mu_i = D \cdot Tr(\rho_A \cdot \Pi_i)$, the probability of outcome i is the normalized temporal bias:

$$P(i) = \mu_i \ / \ \Sigma_j \ \mu_j = Tr(\rho_A \cdot \Pi_i)$$

Since $\rho_A = \Sigma_i |\alpha_i|^2 |A_i\rangle\langle A_i|$ and $Tr(\rho_A \cdot \Pi_\{A_i\}) = |\alpha_i|^2$, this gives $P(i) = |\alpha_i|^2 = |\langle i|\psi\rangle|^2$, which is the Born rule.

We emphasize the structure: rather than deriving the Born rule from first principles, we show that under conditions X1–X4 the assignment of measurement weights reduces to the class of Gleason-type measures, thereby recovering the Born rule within this information-geometric framework. Conditions X1–X4 are geometric properties of the Bures metric; Gleason's theorem is a central, not peripheral, ingredient for the degenerate case.

6. Structural Parallel with the Problem of Time

The Wheeler–DeWitt equation $\hat{H}|\Psi[h_{ij}]\rangle = 0$ encodes the Hamiltonian constraint of canonical quantum gravity. It admits no preferred time parameter. Theorem 1 offers a structural perspective: this equation describes the universe as a whole, without privileging any observer, which corresponds to the complete description G. By Theorem 1, ${}^{O}{}_{o}[\tau] = 0$ for the complete description.

Conversely, Theorem 2 shows that any observer with a partial description that accumulates information experiences an emergent τ with $\mu \neq 0$. The transition from the Wheeler–DeWitt regime to experienced time corresponds to the transition from G to P(o).

We reiterate the scope of this claim. The connection is a structural parallel — both frameworks exhibit atemporality for complete descriptions and emergent time for subsystems — not a constructive embedding. The superspace metric has indefinite signature; the Fisher metric does not. A rigorous identification would require constructing an explicit functor between the respective categories of objects, which we do not do here. This perspective is compatible with, but distinct from, Page–Wootters [14], relational quantum mechanics [15], and Barbour–Bertotti [16]. The distinctive feature is that the atemporality of the complete description is derived from conditions on the information metric under Assumption A1, rather than postulated.

6.1. Role of Initial Entropy and the Thermodynamic Arrow

The typicality argument of Section 3.2 also clarifies the role of low-entropy initial conditions. A globally low-entropy boundary condition can break the symmetry required by Assumption A1 by selecting a preferred direction of information accumulation for a large class of observers. In such a case, the measure on Gr(k,N) is no longer invariant under $\sigma \to -\sigma$, the involution R fails to be measure-preserving, and Theorem 1 no longer applies in its symmetric form.

This framework therefore separates two conceptually distinct issues. First, the atemporal neutrality of the complete description G, captured by A1 and Theorem 1: when A1 holds, the complete description contains no intrinsic temporal bias. Second, the observed thermodynamic arrow of time, which arises precisely when A1 is broken by conditioning, coarse-graining, or special boundary conditions. Low initial entropy is not needed to define a local temporal parameter for any individual observer; but it can explain why large classes of observers share the same macroscopic arrow of time.

Thus the framework offers a structural account of the thermodynamic arrow: it emerges from the breaking of the global informational neutrality of G by cosmological initial data, not from the local dynamics of any individual observer.

7. Experimentally Testable Consequences

The framework makes one prediction that is currently testable and two heuristic consequences that require further derivation.

Prediction 1: Decoherence rate as arctanh of entanglement.

The functional form $\mu(r) = \operatorname{arctanh}(r)$ is proposed as a phenomenological ansatz consistent with the information-theoretic interpretation of τ, rather than a derivation from the formal results of Theorem 2. It assigns to a two-level system entangled with its environment a temporal bias monotonically increasing in $r \in [0,1]$ (the Bloch vector magnitude of the reduced density matrix), vanishing at $r = 0$ (no entanglement, no time) and diverging as $r \to 1$ (maximal entanglement). If this ansatz is accepted, it predicts a decoherence rate $\Gamma(r) = \Gamma_0 \cdot \operatorname{arctanh}(r)$, nonlinear in r and distinguishable from the linear prediction of relational quantum mechanics [15].

The operational definition: prepare a pair of photons in a partially entangled state with concurrence $C = r$ (controllable via wave plates in a BBO crystal setup). Measure the decoherence rate $\Gamma = 1/T_2$ from off-diagonal decay of ρ_A. The prediction differs from the linear form by approximately 10% at $r = 0.5$ and 64% at $r = 0.9$ — both above the 1–2% precision of current entangled-photon experiments [17, 18]. A falsification requires measuring Γ at five or more values of r in [0.3, 0.9] and fitting to both functional forms.

Heuristic Consequence 2: Electroweak mixing angle at the GUT scale.

The information-geometric structure in four dimensions, based on the self-duality of SO(4) and the symmetry breaking $SU(2)_R \to U(1)_Y$, is compatible with $\sin^2(\theta_W)|_{GUT} = 1/4$ at the unification scale, distinguishable from the SU(5) prediction of 3/8. This should be regarded as a heuristic consequence of the framework; the derivation from the information-geometric axioms is

not completed in this paper and requires further work before experimental comparison is meaningful.

Heuristic Consequence 3: Stability of the electroweak vacuum.

The condition $R_{Fisher} = 0$ at the electroweak vacuum — identified with the Ricci-flat condition for the statistical manifold — implies complete stability of the vacuum, in contrast to the Standard Model prediction of metastability [19]. This consequence also requires further derivation and should not yet be treated as a quantitative prediction.

8. Open Problems and Future Directions

Derivation of the Principle of Global Informational Neutrality. A central open problem is to determine whether the Principle of Global Informational Neutrality, and with it Assumption A1, can be derived from more primitive assumptions such as unitarity, typicality of subsystems, or cosmological boundary conditions. If such a derivation is possible, Assumption A1 would become a theorem rather than a structural postulate, and the atemporality result of Theorem 1 would follow from the dynamical laws of the theory. If A1 cannot be so derived, a precise characterization of the class of physical systems and cosmological scenarios for which it holds remains an important open problem.

The derivation of SU(3). A complete algebraic derivation of the gauge group structure $SU(3) \times SU(2) \times U(1)$ from the information-geometric structure in four dimensions is work in progress.

Fermion masses and the Yukawa sector. The framework does not yet account for fermion masses. Deriving the Yukawa couplings from the geometry of Fisher manifolds remains an open problem.

The cosmological constant. The smallness of the observed cosmological constant relative to the vacuum energy density is not addressed in this paper and represents a significant challenge for any information-geometric approach to fundamental physics.

9. Conclusions

We have presented an information-geometric framework in which three fundamental features of physics arise from a common structure. The key results are:

(1) Under Assumption A1 (reflection symmetry of the space of partial descriptions), any complete description is atemporal: the expected temporal parameter over all observers vanishes exactly (Theorem 1). Assumption A1 is an explicit additional hypothesis, not derived from C1–C3.

(2) A continuous temporal parameter τ emerges for any observer with a partial description that accumulates information through successive interactions (Theorem 2).

The arrow of time is the statistical bias $\mu = {}^{O}[\sigma]$ of the binary anomaly variable. Time is the bias in the observer's learning process relative to the complete description G.

(3) Under conditions X1–X4, the assignment of measurement weights reduces to the class of Gleason-type measures, recovering the Born rule $P(i) = Tr(\rho\Pi_i)$ within this framework (Theorem 3). Gleason's theorem is a central ingredient for the degenerate case.

The framework establishes a structural parallel between the atemporality of the Wheeler–DeWitt equation and the emergence of time for internal observers, without claiming a rigorous embedding. The connection to canonical quantum gravity and the heuristic predictions of Section 7.2–3 define the open problems for future work. Prediction 1 (decoherence rate as arctanh of entanglement) is experimentally distinguishable from existing alternatives with current technology. The framework is intended as a structural unification rather than a complete dynamical theory.

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Appendix: Verification that C1–C3 Correspond Exactly to the Hypotheses of Petz's Theorem

We verify explicitly that the three conditions satisfy exact quantum analogues that are precisely the hypotheses of Petz's characterization theorem [8].

C1 (Quantum case). Statistical sufficiency corresponds to CPTP monotonicity: a metric g satisfies C1 if and only if $g_{\Lambda(\rho)}(\Lambda(A), \Lambda(A)) \leq g_\rho(A, A)$ for all CPTP maps Λ. This is precisely the defining condition in Petz [8, Theorem 1].

C2 (Quantum case). Reparametrization invariance corresponds to unitary invariance: $g_{U\rho U^\dagger}(UAU^\dagger, UBU^\dagger) = g_\rho(A, B)$ for all unitary U. This is built into the Petz construction via the operators L_ρ and R_ρ, which transform covariantly under unitary conjugation.

C3 (Quantum case). Statistical locality requires additivity on product states. For the Petz family, each term in the Löwner integral factorizes over tensor products because $L_{\rho_A \otimes \rho_B} = L_{\rho_A} \otimes L_{\rho_B}$ and $R_{\rho_A \otimes \rho_B} = R_{\rho_A} \otimes R_{\rho_B}$. The cross terms vanish because tangent vectors at product states have zero cross-system components.

The three mappings are exact: satisfying C1, C2, C3 in the quantum setting is equivalent to satisfying the hypotheses of Petz's theorem, and conversely.

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